

VLSI Lab

Laboratory report submitted for the partial fulfillment
of the requirements for the degree of

Bachelor of Technology
in
Electronics and Communication Engineering

by

Name :- ARPIT RAJ Roll No. :- 24UEC011

Course Coordinator
Dr. Sandeep Saini



Department of Electronics and Communication Engineering
The LNM Institute of Information Technology, Jaipur

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Chapter 1

Experiment - 2

1.1 Name of the Experiment

To introduce and study the electrical characteristics of NMOS and PMOS transistors.

1.2 Software/ Apparatus/ Components Used

1. LT-SPICE software installed on a computer
2. NMOS transistor model
3. PMOS transistor model
4. Power supply setup (virtual in LT-SPICE)
5. Resistors (values as per circuit design)
6. Signal source (voltage and current sources)

1.3 Theory

NMOS and PMOS are two types of MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors) widely used in analog and digital circuits. MOSFETs operate in three regions: cutoff, triode, and saturation. The behavior of NMOS and PMOS is determined by the applied gate-to-source voltage (V_{GS}) and drain-to-source voltage (V_{DS}).

NMOS Characteristics: The NMOS transistor conducts when the gate voltage is higher than the source voltage by a threshold voltage (V_{th}). The source is usually connected to ground.

PMOS Characteristics: The PMOS transistor conducts when the gate voltage is lower than the source voltage by a threshold voltage. The source is typically connected to a positive voltage.

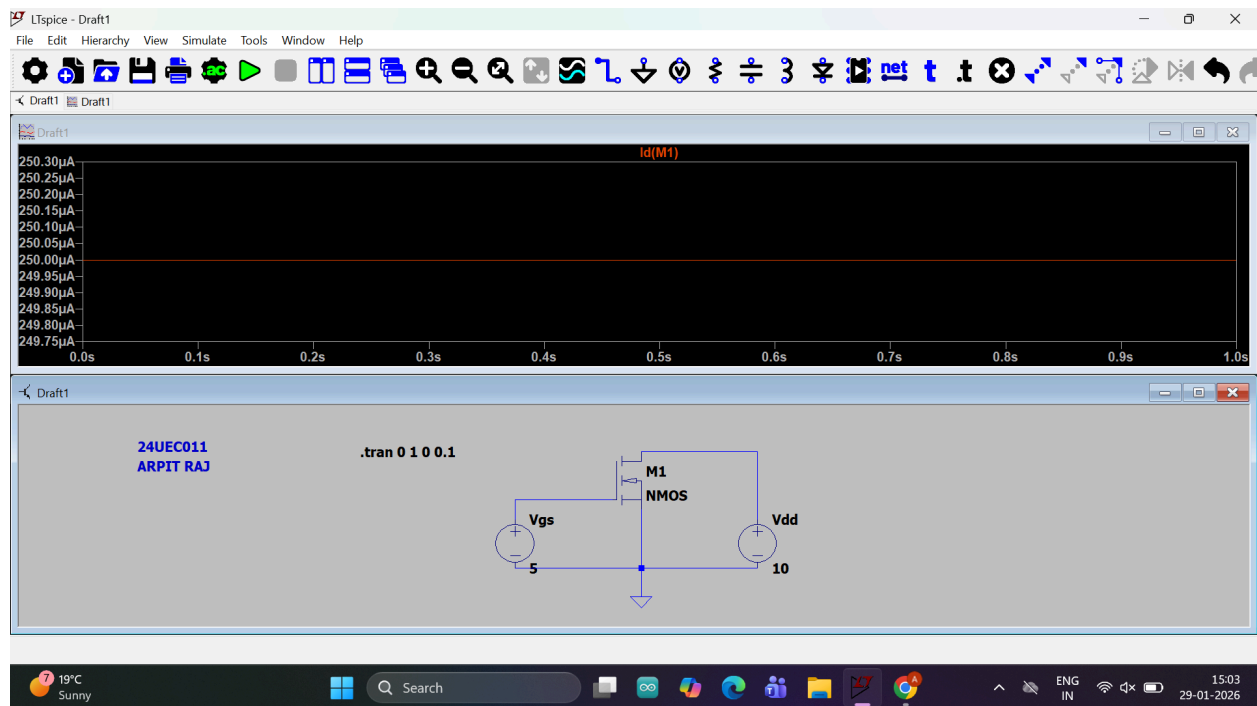
1.4 Simulation and Results

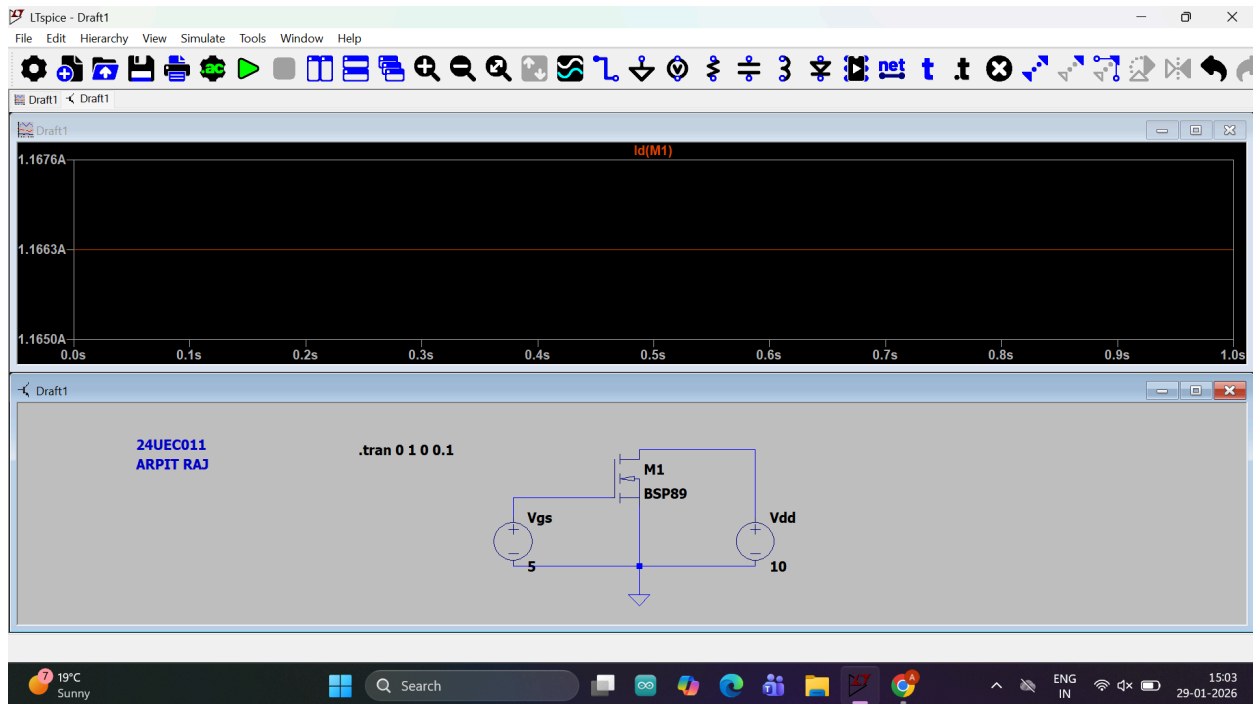
1.4.1 NMOS Characterization:

a) Steps used to draw the Circuit

- a. Place an NMOS transistor on the schematic.
- b. Connect $V_{gs} = 5V$ and $V_{ds} = 10V$.
- c. Setup transient analysis to run for 1 second.
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- d. Run the simulation and observe the output characteristics.
- e. This shows almost negligible current even though enough voltage was provided at V_{gs} and V_{ds} .
- f. This is due to the wrong selection of NMOS.
- g. Right click on the NMOS and select “Pick new MOFET”, and select BSP89.
- h. Run the simulations again.
- i. Now sufficient current is obtained.

b) Schematic Diagram and Simulation and Results

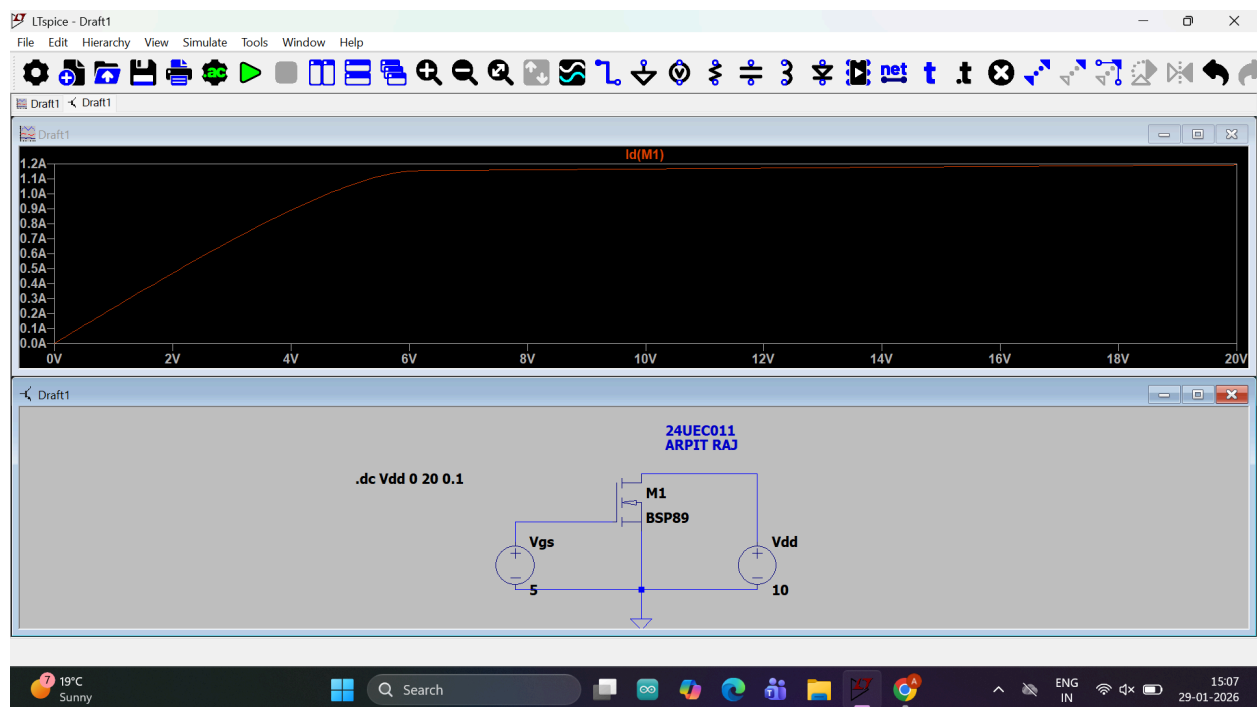
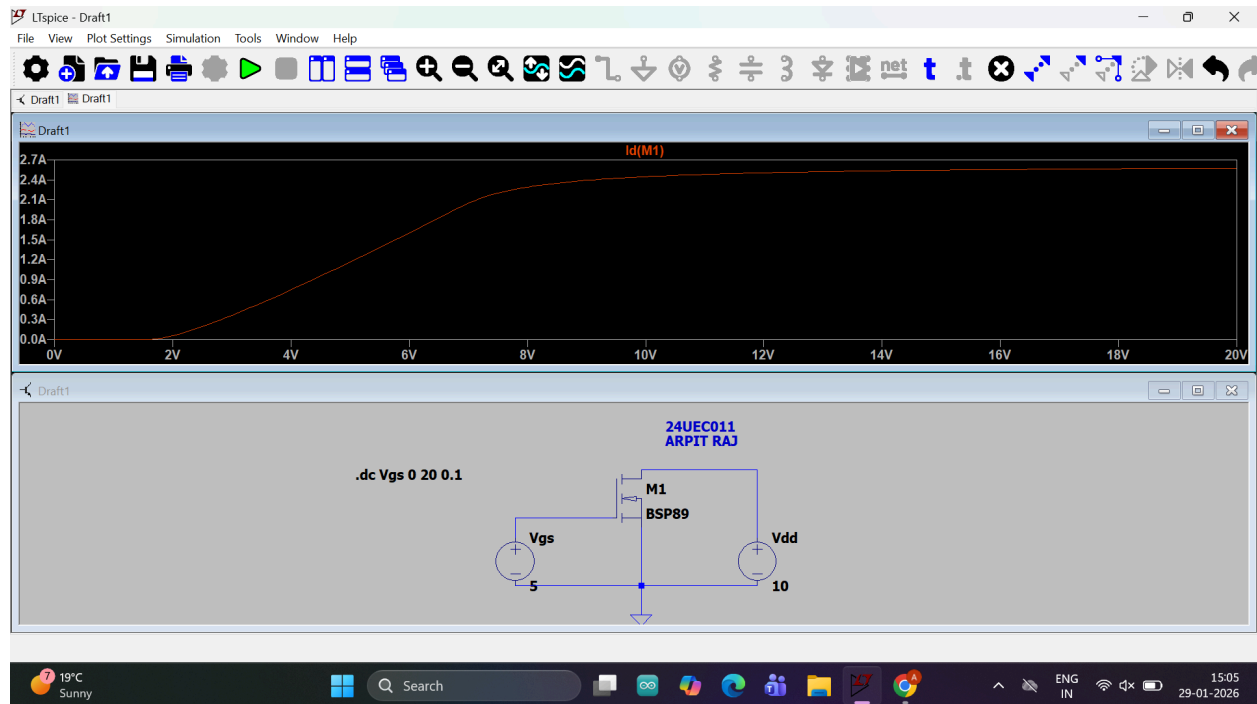




1.4.2 I-V characteristics of NMOS

a) Steps used to draw the Circuit

1. We'll use the DC sweep analysis for I-V characteristics of an NMOS.
2. Set Vgs for DC sweep analysis with the following settings `.dc Vgs 0 20 .1`
3. Run the simulation and observe the current at the drain.
4. We can see that the drain current starts increasing at Vgs of 1.9 and above. And then it saturates.
5. Now let us sweep with respect to the Vds.
6. Keep Vgs = 5V and apply the DC sweep at Vds with the following settings: `.dc Vds 0 20 .1`
7. Observe the drain current.
8. In the next step, let us plot multiple I-V lines for different values of Vds.
9. This can be done in two different ways:
 1. Using multiple sources for DC sweep
 2. Using parameters for one of the sources and sweep at the other.



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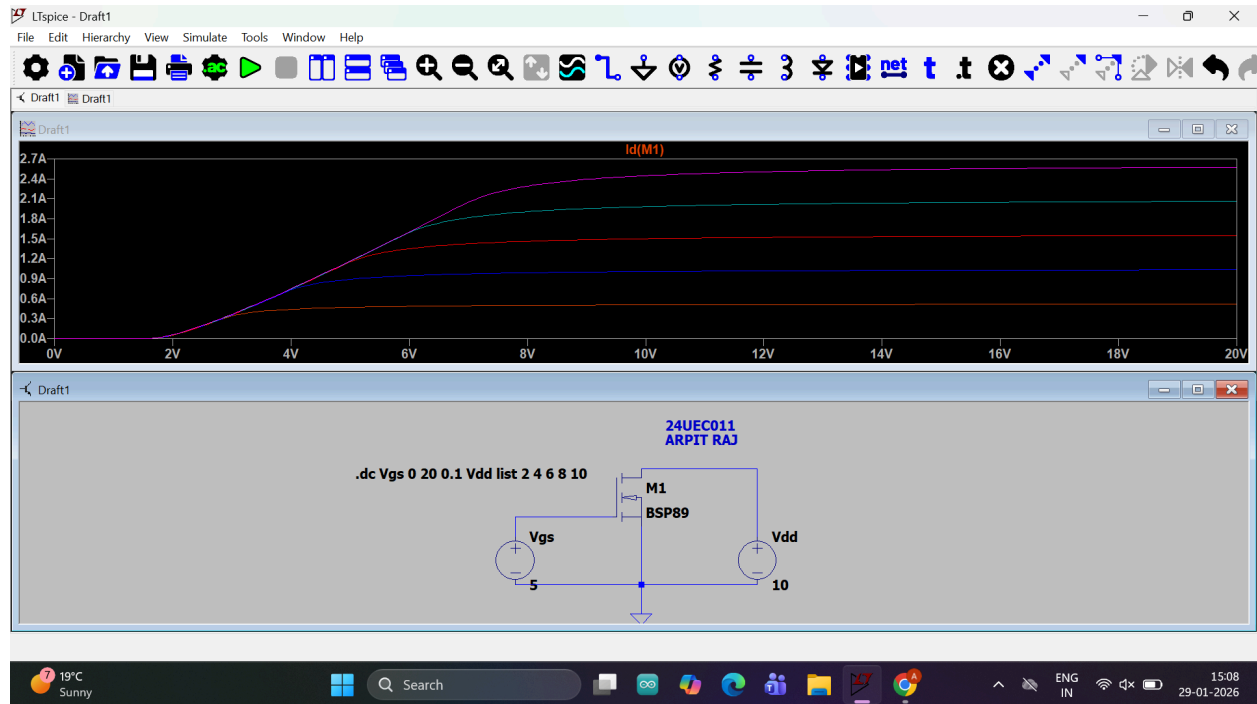
1.4.3 I-V characteristics of NMOS using multiple sources for DC sweep a) Steps used to draw the Circuit

1. Apply the DC sweep at V_{gs} with the following settings: `.dc Vgs 0 20 .1`
2. Apply the DC sweep at V_{ds} with the following settings: `.dc Vds 0 20 2`. This means that we are

changing the value of V_{ds} with a difference of 2 volts and we expect 10 different curves.

3. Run the simulation and observe the I_d curves.

b) Schematic Diagram and Simulation and Results



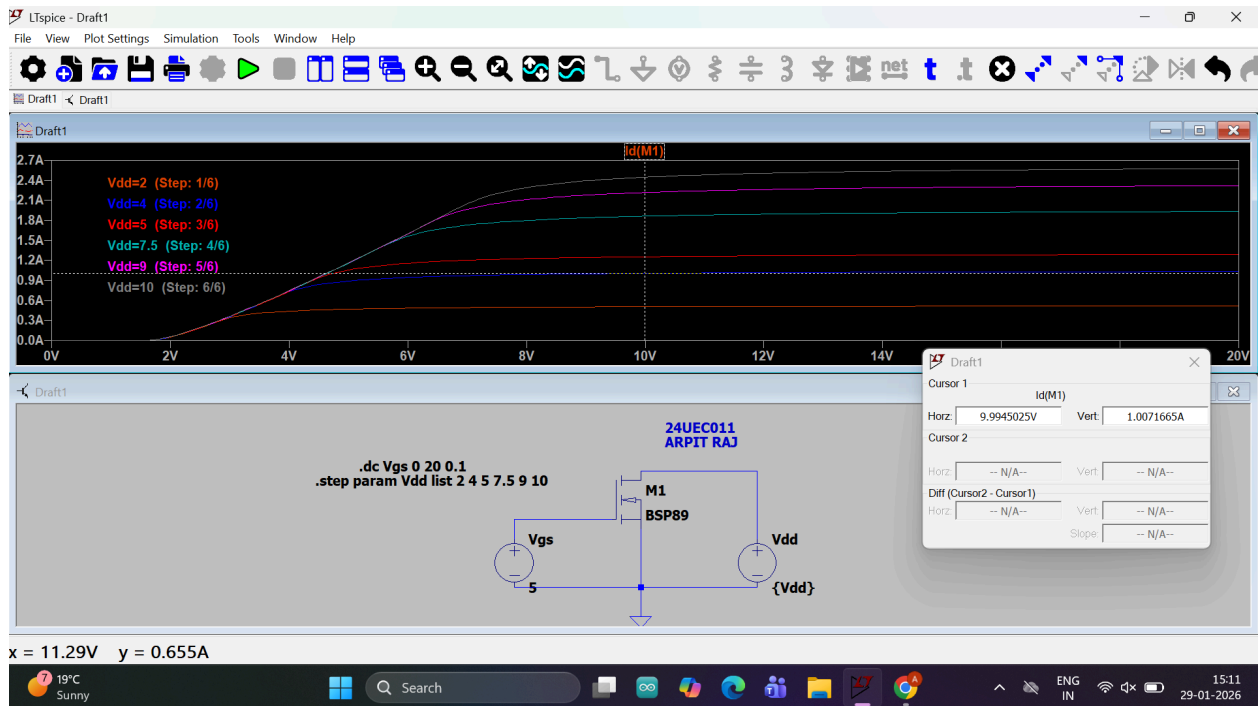
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1.4.4 I-V characteristics of NMOS using parameters for one of the sources and sweep at the other.

a) Steps used to draw the Circuit

1. Apply the DC sweep at V_{gs} with the following settings: `.dc Vgs 0 20 .1`
2. Right click on the voltage source V_{ds} and give DC values as $\{V_{ds}\}$. This would set this voltage source as parameter V_{ds}
3. Select SPICE directives (black t icon) from the top.
4. In the pop-up window, give command `.step param Vds 2 4 6 8 10`. This means that we are giving a list of values to parameter V_{ds} and the values are 2, 4, 6, 8, and 10.
5. Run the simulation and observe I-V characteristics.

b) Schematic Diagram and Simulation and Results

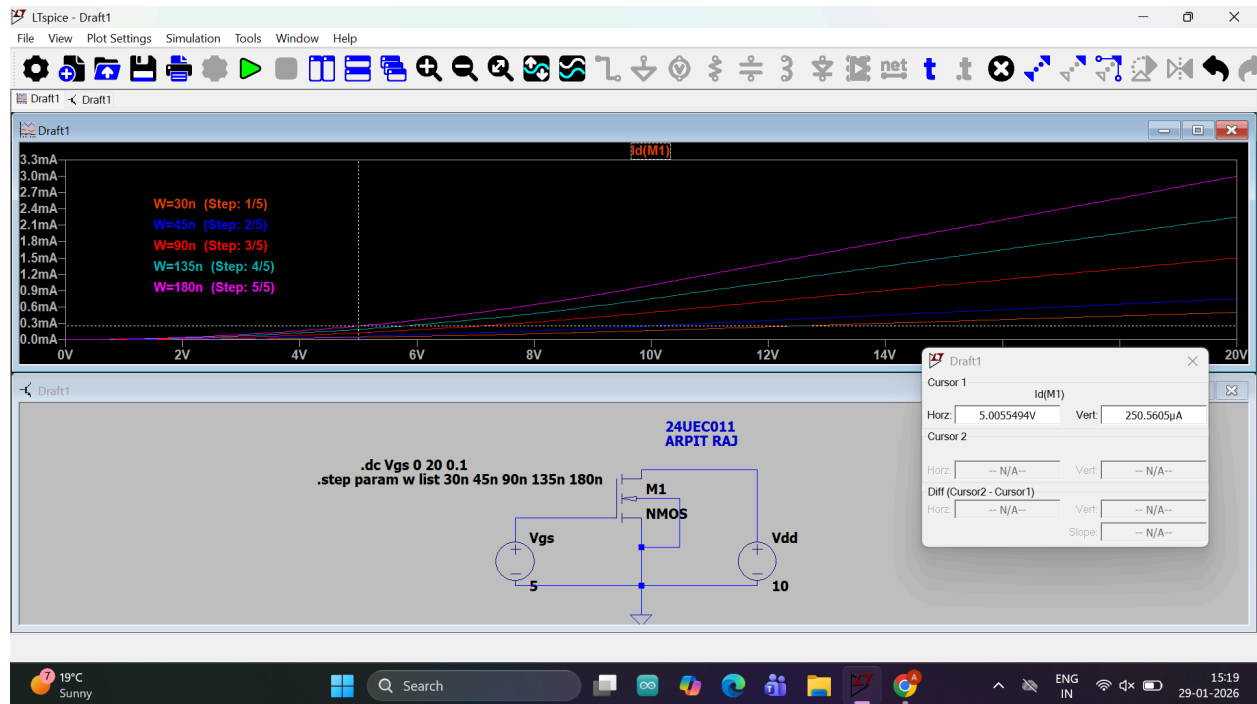
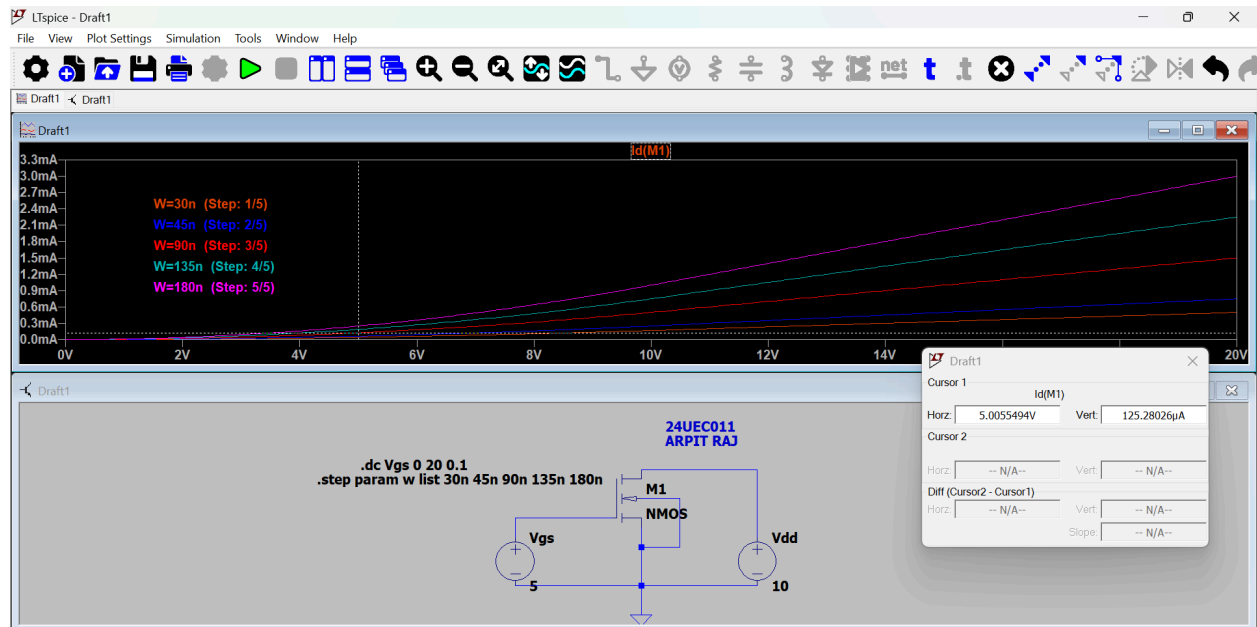


1.4.5 Effect of MOS parameters on I-V characteristics

a) Steps used to draw the Circuit

1. Draw a new schematic which is similar to the previous ones but take a new NMOS from the component library. Select nmos4 from the library.
2. Ensure that the body is connected to the ground.
3. Take $V_{ds} = 10V$ and sweep V_{gs} from 0 to 20 volts.
4. Right click the NMOS and select length = 180n and width = 90n. Do not change anything else.
5. Run the simulations and observe the value of I_D at $V_{gs} = 2V$ and $4V$
6. Change the value of length to 45n and observe the value of I_D at $V_{gs} = 2V$ and $4V$

b) Schematic Diagram and Simulation and Results



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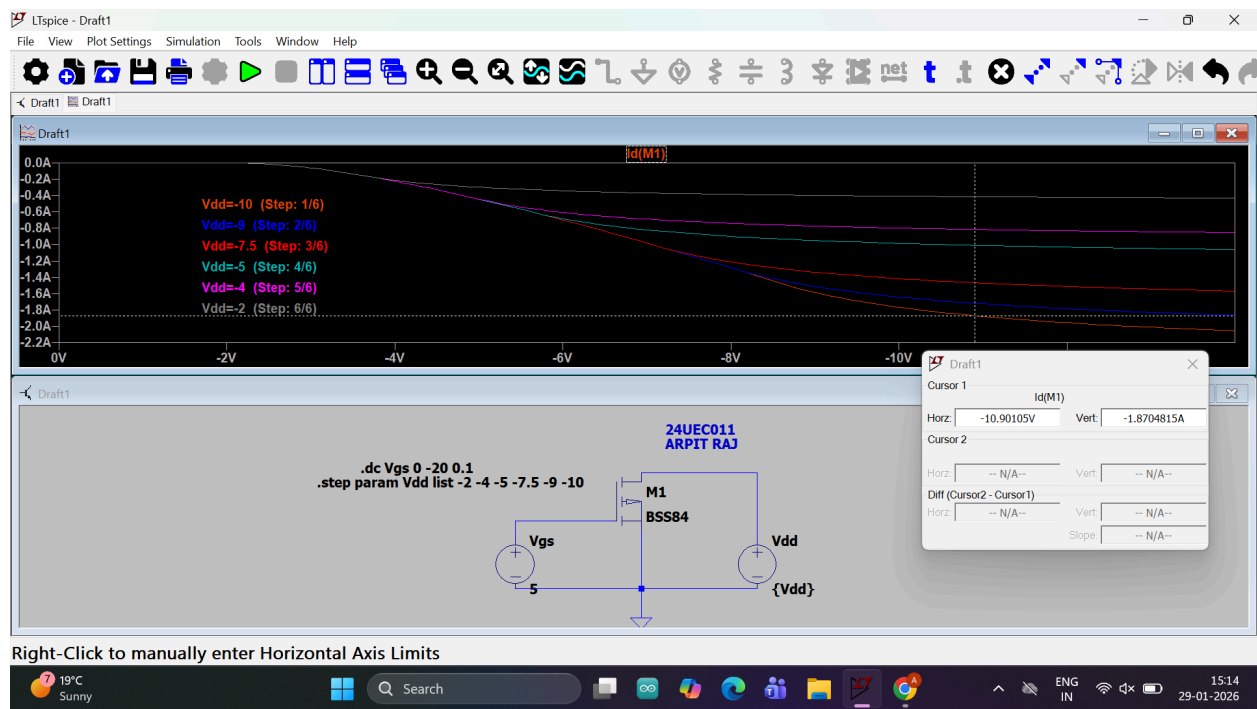
LENGTH (90n) WIDTH (180n)	2V 80uA	4V 320uA
LENGTH (45n) WIDTH (180n)	2V 160uA	4V 640uA

1.4.6 I-V characteristics of PMOS

a) Steps used to draw the Circuit

1. Design the circuit with proper biasing of PMOS.
2. Take BSS84 PMOS.
3. V_{gs} and V_{ds} will be negative.
4. Set sweep points for V_{gs} and V_{ds}
5. Run the simulations and observe I_D .

b) Schematic Diagram and Simulation and Results



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1.5 Summary

NMOS (N-channel MOSFET) and PMOS (P-channel MOSFET) are two types of MOSFETs used in digital and analog circuits.

Feature NMOS (N-channel MOSFET) PMOS (P-channel MOSFET) Carrier Type Electrons

(faster) Holes (slower)

Turn-On Condition

(holes have lower mobility)

Switching Speed

Requires **positive** gate-to-source voltage (VGS)

Requires **negative** gate-to-source voltage (VGS)

Faster (electrons have higher mobility) Slower

ON Resistance Lower Higher

(CMOS circuits use NMOS for pull-down)

Used in pull-up networks in CMOS circuits

Symbol Notation

Arrow pointing **out** of the source Arrow pointing
into the source

Usage Preferred in high-speed applications

The **threshold voltage** (V_{TH} or V_{TH}) of a MOSFET is the minimum gate-to-source voltage (VGS) required to create an inversion layer in the channel, allowing current to flow between the drain and source.

- **For NMOS:** V_{TH} is positive. The transistor turns **ON** when $V_{GS} > V_{TH}$
- **For PMOS:** V_{TH} is negative. The transistor turns **ON** when $V_{GS} < V_{TH}$